

Figure 4.1 Work done by a force

Here, the product $\vec{F} \cdot d\vec{r}$ is a *scalar product (or dot product)*. The scalar product of two vectors is a scalar (refer section 2.5.1). Thus, work done is a scalar quantity. It has only magnitude and no direction. In SI system, unit of work done is N m (or) joule (J). Its dimensional formula is $[ML^2T^{-2}]$.

The equation (4.1) is,

$$W = F\, dr \cos\theta \qquad (4.2)$$

which can be realised using Figure 4.2 (as $\vec{a} \cdot \vec{b} = ab\cos\theta$) where, θ is the angle between applied force and the displacement of the body.

The work done by the force depends on the force (F), displacement (dr) and the angle (θ) between them.

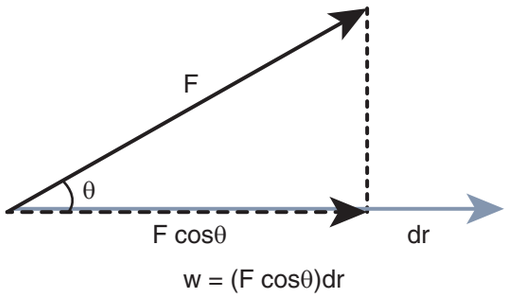


Figure 4.2 Calculating work done.

Work done is zero in the following cases.

(i) **When the force is zero (F = 0).** For example, a body moving on a horizontal smooth frictionless surface will continue to do so as no force (not even friction) is acting along the plane. (This is an ideal situation.)

(ii) **When the displacement is zero (dr = 0).** For example, when force is applied on a rigid wall it does not produce any displacement. Hence, the work done is zero as shown in Figure. 4.3(a).

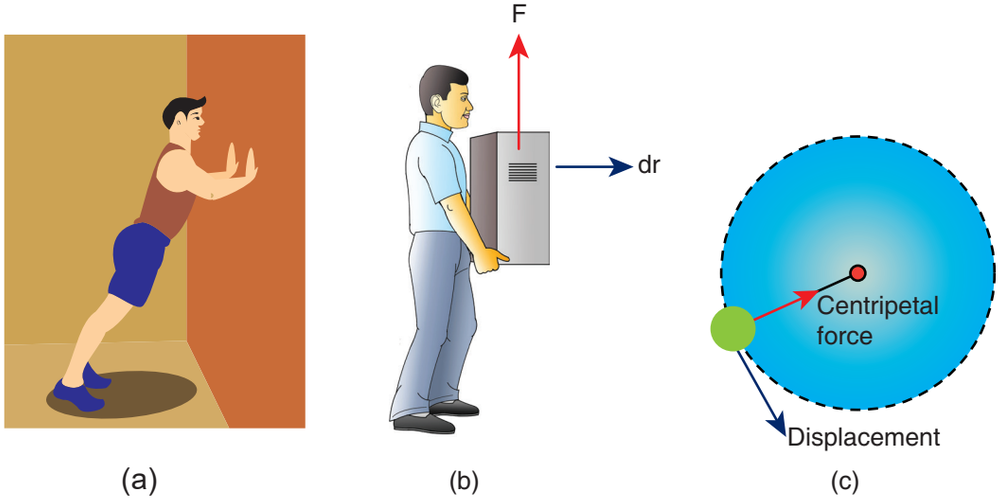


Figure 4.3 Different cases of zero work done

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